

On the Repeatability of the Apex–Atlas Apparatus

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3. Methods

3.1 The apparatus consists of a 30 mm copper plate clamped between two ceramic supports, instrumented with three K-type thermocouples spaced at 10 mm intervals along the plate's centerline.

3.2 An *Operator* in this study is defined as any individual who performs the full clamp-instrument-record cycle without external assistance, including calibration of the cold-junction reference and visual verification of the seal. Operators were trained for two hours and then tested for inter-operator agreement ($\kappa = 0.83$).

4. Findings

4.1 Across twelve trials run by four operators, the standard deviation of measured thermal conductivity was 1.7% of the mean, a substantial improvement over the 6.3% reported for the prior apparatus. The improvement is attributable principally to the cold-junction protocol introduced in §3.2.

4.2 Variance attributable to operator identity was 0.4% (95% CI 0.2–0.7%), consistent with the inter-operator agreement reported in §3.2.